



Scientific machine learning

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Abstract Scientific Machine Learning (SciML) is an emerging interdisciplinary field that integrates the strengths of scientific computing and machine learning to address complex modeling, simulation, and inference tasks in the natural and engineering sciences. This paper provides a concise overview of the foundational paradigms underlying SciML, where we highlight the complementary roles of physics-based models and data-driven methods. We discuss the core challenges, including data scarcity, physical consistency, high dimensionality, and computational cost and then introduce key methodological advances such as physics-informed neural networks, operator learning, hybrid modeling, and probabilistic approaches, each designed to address specific limitations of traditional methods. The techniques are illustrated through applications in fluid dynamics and turbulence modeling, automation and control, energy management, and sustainable mobility, demonstrating how SciML enables more accurate, efficient, and interpretable solutions. Finally, we outline open problems and future directions, and emphasize the need for theoretical understanding, scalable algorithms, and interdisciplinary collaboration.

Keywords Scientific computing · Machine learning · Physics-informed learning · Operator learning

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1 Introduction

Over the past seventy years, numerical analysis has developed into a mature discipline at the heart of computational science. From the early finite difference schemes of the 1950s to the sophisticated finite element and spectral methods of today, its focus has been on stability, convergence, and error control, underpinned by rigorous mathematical theory. Nowadays we merely speak about scientific computing, enabling the simulation and analysis of complex physical, biological, and engineering systems through mathematical models, typically formulated as differential equations. Meanwhile, the past two decades have seen the explosive rise of machine learning (ML). Initially a branch of artificial intelligence rooted in statistics and optimization, ML has evolved into a powerful framework for extracting patterns from large datasets. Deep learning, in particular, has achieved spectacular successes in domains where explicit physical modelling is either impossible or impractical, offering powerful tools for pattern recognition, prediction, and representation learning across diverse application areas.

Scientific Machine Learning (SciML) sits at the intersection of the above two paradigms [1]. It seeks to integrate the rigor and interpretability of physics-based models and numerical analysis techniques with the flexibility and scalability of machine learning [2]. This synthesis is not merely a juxtaposition of methods, but a conceptual shift: from purely mechanistic modelling to hybrid approaches that exploit both data and structure. SciML addresses fundamental challenges in modern computational science, such as modelling systems with incomplete knowledge, accelerating simulations, and solving high-dimensional inverse problems. Historically, numerical analysis and scientific computing have emphasized stability, convergence, and error control, while machine learning has prioritized generalization, expressivity, and data efficiency. Reconciling these perspectives requires new mathematical frameworks and algorithmic innovations. SciML is thus not only a computational endeavour but also a fertile ground for new mathematical theories. For mathematicians, SciML offers both opportunities and challenges. It raises deep theoretical questions on the approximation, stability, and convergence of physics-informed learning methods; invites the design of algorithms that preserve structure such as conservation laws and symmetries; and connects to long-standing interests in reduced-order modelling, operator theory, and inverse problems. Moreover, it provides new arenas for mathematics to influence fields as diverse as climate science, materials design, and medicine—while bringing back novel computational problems and perspectives into the mathematical community. In this sense, SciML is not just an application area, but a two-way street between mathematics and the sciences, with potential for transformative impact on both.

This paper provides a concise overview of the principles and challenges of SciML, followed by a focused discussion of methodological advances and application areas. Section 2 introduces the foundational paradigms and outlines the core difficulties in merging them. Section 3 presents key algorithmic developments, including physics-informed neural networks and operator learning. Section 4 explores applications in fluid dynamics and turbulence modeling, automation and control, energy management, and sustainable mobility. Section 5 discusses open questions, future directions,

and concludes with an outlook on the evolving role of SciML in computational science.

2 Foundations and challenges of scientific machine learning

Scientific Machine Learning (SciML) lies at the intersection of two well-established disciplines: scientific computing and machine learning. This section provides an overview of the foundational principles of both paradigms and outlines the key challenges that arise in their synthesis.

2.1 The scientific computing paradigm

Scientific computing is concerned with the numerical solution of mathematical models that describe physical, biological, and engineered systems. These models are typically formulated as systems of differential equations, integral equations, or variational problems, derived from first principles such as conservation laws, constitutive relations, or empirical observations.

A central role is played by partial differential equations (PDEs), which govern a wide range of phenomena including fluid flow, heat transfer, electromagnetism, and elasticity [3]. The general form of a PDE-based model can be written as

$$\mathcal{L}(u; \theta) = 0 \quad \text{in } \Omega, \quad \mathcal{B}(u) = 0 \quad \text{on } \partial\Omega,$$

where $\mathcal{L}$ is a differential operator parameterized by physical parameters θ , $\mathcal{B}$ denotes boundary conditions, and u is the unknown solution defined on a spatial domain Ω .

To solve such models numerically, one typically discretizes the domain using finite difference, finite element, or spectral methods, leading to large-scale algebraic systems. The design of these methods is guided by principles of consistency, stability, and convergence (including, e.g., CFL conditions).

In addition to forward simulation, scientific computing also addresses inverse problems, where one seeks to infer unknown parameters or inputs from observed data [4, 5]. These problems are often ill-posed and require regularization techniques and uncertainty quantification (UQ, [6]) to ensure meaningful solutions. Bayesian formulations offer a probabilistic framework for incorporating prior knowledge and quantifying uncertainty [7].

Despite its successes, traditional scientific computing faces several limitations:

- **Model complexity:** High-fidelity models can be computationally expensive, especially in high dimensions or when involving multiscale phenomena.
- **Incomplete knowledge:** Governing equations may be only partially known or involve empirical closures.
- **Data integration:** Incorporating experimental or observational data into simulations is challenging.

These challenges motivate the integration of data-driven machine learning techniques, which offer complementary strengths in approximation, data assimilation, and model discovery.

2.2 The machine learning paradigm

Machine learning (ML) is a branch of artificial intelligence concerned with the development of algorithms that improve their performance when more information becomes available, typically in the form of data. At its core, ML seeks to approximate functions or distributions based on observed input-output pairs, often without explicit knowledge of the underlying generative process [8, 9].

In supervised learning, the goal is to learn a mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ from a dataset $\{(x_i, y_i)\}_{i=1}^N$, where $x_i \in \mathcal{X}$ are inputs and $y_i \in \mathcal{Y}$ are outputs. The learning process involves minimizing a loss function $\mathcal{L}(f(x), y)$, typically using gradient-based optimization. In unsupervised learning, the objective is to uncover structure in unlabeled data, such as clusters, manifolds, or latent variables.

Neural networks, particularly deep neural networks, have emerged as powerful function approximators [8]. A feedforward neural network with parameters θ defines a composition of affine transformations and nonlinearities:

$$f_{\theta}(x) = W_L \sigma(W_{L-1} \cdots \sigma(W_1 x + b_1) \cdots + b_{L-1}) + b_L,$$

where σ is a nonlinear activation function (e.g., ReLU, tanh), and $\theta = \{W_{\ell}, b_{\ell}\}$ are learnable parameters, called “weights” and “biases”. The universal approximation theorem [10] guarantees that such networks can approximate continuous functions arbitrarily well, given sufficient width (or depth [11]).

A key concept in ML is *generalization*—the ability of a model to perform well on unseen data. This is influenced by the model’s capacity, the amount and quality of training data, and the choice of regularization. Overfitting occurs when a model captures noise or spurious patterns in the training data, leading to poor predictive performance.

Another central idea is *inductive bias*, which refers to the assumptions a learning algorithm makes to generalize beyond the training data. In scientific applications, incorporating domain knowledge as inductive bias—such as symmetries [12], conservation laws [13], or known dynamics [14]—can significantly improve learning efficiency and robustness.

While ML has achieved impressive results in areas such as computer vision, natural language processing, and reinforcement learning, its application to scientific problems poses unique challenges. These include the need for physical consistency, interpretability, and the integration of sparse or noisy data with complex models.

2.3 Core challenges in SciML

The integration of machine learning with scientific computing introduces a range of conceptual and practical challenges that go beyond those encountered in traditional data-driven or model-based approaches. These challenges arise from the need to

respect physical laws, operate under data constraints, and ensure computational efficiency and robustness.

2.3.1 Data scarcity and noise

Unlike many commercial applications of machine learning, scientific domains often suffer from limited and noisy data. High-fidelity simulations or experiments can be expensive or time-consuming, and measurements may be sparse in space and time. This scarcity requires models and algorithms that can generalize from small datasets and incorporate prior knowledge effectively, for example, through multi-fidelity hierarchies [15–17].

2.3.2 Physical consistency and interpretability

Scientific models are expected to obey known physical laws, such as conservation of mass, energy, or momentum. Standard machine learning models, however, are typically agnostic to such constraints and may produce physically implausible results. Ensuring that learned models respect such physical laws is a central concern in SciML [18, 19]. Moreover, interpretability is crucial for scientific insight and trust, motivating the development of models whose structure and behavior can be understood and analyzed [20, 21].

2.3.3 High dimensionality and stiffness

Many scientific systems are governed by high-dimensional PDEs or involve multi-scale dynamics with stiff behavior [22, 23]. Learning in such settings is challenging due to the curse of dimensionality and the need for numerical stability. Efficient representations [24, 25], such as low-rank approximations or hierarchical models, are often required to make learning tractable.

2.3.4 Computational cost and scalability

Training modern machine learning models can be computationally intensive, especially when combined with large-scale simulations or optimization loops. Scientific applications often demand real-time or many-query performance, such as in uncertainty quantification, control, or design [6]. This means it is important to develop surrogate models, reduced-order methods, well-designed software, and parallel algorithms that can scale to high-performance computing environments [26, 27].

2.3.5 Integration of models and data

A defining feature of SciML is the need to integrate mechanistic models with observational data. This includes learning unknown components of a model [28], correcting model bias [29], or assimilating data into simulations. Achieving this integration requires new formulations that blend differential equations with statistical inference and machine learning architectures. It also concerns learning entirely new algorithms

[30, 31], which is possible with the combination of general algorithm architectures from numerical analysis and the data-driven refinement from machine learning [32, 33]. Transferability of methods across problems and domains is a related challenge. In many cases, we observe that each problem requires both different physics and different data, and thus calls for new combinations. Thus, it seems challenging to devise standardized approaches that work well across many problems. These challenges motivate the development of novel methodologies that combine the strengths of both paradigms.

3 Methodological advances in SciML

Scientific Machine Learning has given rise to a variety of novel methods that integrate structure and constraints of scientific models into machine learning architectures. These methods can in part overcome the limitations of purely data-driven or purely mechanistic approaches, e.g., by embedding physical knowledge into learning algorithms, improving generalization [34], interpretability, and efficiency. This section presents several prominent methodological advances, including physics-informed neural networks, operator learning, hybrid modeling, and probabilistic approaches.

3.1 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs) are a class of neural networks designed to solve differential equations by incorporating physical laws directly into the training process [13, 35–38]. Introduced by Lagaris [35] and greatly extended and popularized by Raissi, Perdikaris, and Karniadakis [39], PINNs embed the governing equations of a system into the loss function of a neural network, thereby guiding the learning process with known physics. Consider a PDE of the form

$$\mathcal{L}(u(x, t)) = 0, \quad (x, t) \in \Omega \times [0, T],$$

with appropriate initial and boundary conditions. In a PINN framework, a neural network $u_\theta(x, t)$ is trained to approximate the solution $u(x, t)$. The loss function typically consists of multiple components:

$$\mathcal{L}_{\text{PINN}} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{PDE}} + \mathcal{L}_{\text{BC}},$$

where $\mathcal{L}_{\text{data}}$ penalizes deviations from observed data, $\mathcal{L}_{\text{PDE}}$ enforces the differential equation (via automatic differentiation), and $\mathcal{L}_{\text{BC}}$ enforces boundary and initial conditions. The idea of integrating PDE knowledge during training offers several advantages.

- The approach can solve forward and inverse problems without requiring a lot of labeled data for the solution (only boundary and initial data is required).

- Neural networks offer a mesh-free ansatz and can thus handle irregular geometries and domains.
- Incorporating physical constraints during training can improve generalization and robustness.

Notable challenges have been discussed at length in the literature, too.

- Training can be difficult for stiff or high-dimensional problems, often requiring careful loss balancing and network design [40].
- Convergence guarantees are limited, and theoretical understanding is still developing.
- Computational cost can be high, especially for long-time integration or complex geometries.
- The literature for PINN-based learning approaches is very diverse and suffers from weak baselines [41].

Recent developments include adaptive weighting strategies, domain decomposition (e.g., FBPINNS [38]), and the use of transfer learning [42] and curriculum learning to improve training efficiency. PINNs have been successfully applied to problems in fluid dynamics, solid mechanics, and electromagnetism [43].

3.2 Operator learning and neural operators

While traditional machine learning models focus on learning finite-dimensional mappings, many scientific problems involve learning operators—mappings between infinite-dimensional function spaces. For example, solving a PDE corresponds to learning a solution operator that maps initial or boundary conditions (or source terms) to the solution function. Operator learning aims to approximate such mappings directly, bypassing the need to solve the PDE repeatedly for different inputs.

Let $\mathcal{G} : \mathcal{X} \rightarrow \mathcal{Y}$ be an operator that maps a function $a(x) \in \mathcal{X}$ (e.g., a coefficient field or forcing term) to a solution $u(x) \in \mathcal{Y}$. The goal is to learn an approximation $\mathcal{G}_\theta$ from data $\{(a_i, u_i)\}_{i=1}^N$, where each pair represents a realization of the input-output functions. Several architectures have been proposed for operator learning [44], most prominently Deep Operator Networks (DeepONets, [45, 46]) and Fourier Neural Operators (FNOs, [47, 48]). A more scientific computing oriented approach is possible by approximating a known operator, e.g. the Laplace operator on a Riemannian manifold [20, 49–53], the Koopman operator of dynamical systems [54–56], or the lift-and-learn framework [57].

In the lifetime of a product family many design variations are simulated, e.g., due to an evolution of the product design. Thereby, lengthy simulation times can become quickly a bottleneck in the overall design process. Given that a number of similar geometries with only slight design modifications are simulated, these applications are well suited for ML. That is, a set of corresponding potential design variations is simulated upfront to the design process and a corresponding ML-model is trained. Thereby the variation of the design must not necessarily explicitly parameterized but alternative geometries be created via generative Machine Learning (ML) models, e.g., via (variational) autoencoders, e.g., [58], or diffusion models, e.g., [59]. Often

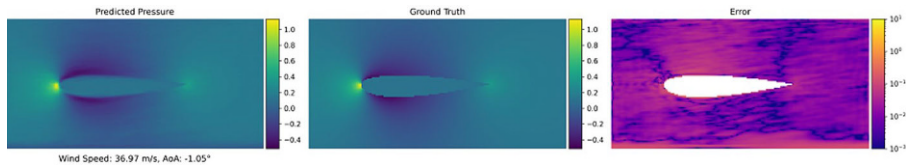


Fig. 1 Prediction of a pressure distribution around an airfoil (data from [68]) with a U-Net architecture

such an approach might not be viable from the corresponding computational budget spend, i.e., many more simulations might be required to train the corresponding ML model than actually the ML model is evaluated [60]. However, the gain in the time-to-market has often a much higher value than the computational costs.

One of the first ML concepts proposed in the context of Computational Fluid Dynamics (CFD) has been the U-net architecture, ([61, 62]; see Fig. 1) but most current architectures use Transformer-type architectures, e.g., [63, 64]. With the ability to effectively train corresponding models, ideas for pre-trained models or even foundation models emerge. However despite bold claims, the effectively and realism of corresponding approaches is not clear [65]. Key challenges for research is mostly the access to corresponding datasets, since most available datasets are oversimplified [66–71] and realistic cases for industrial applications are rather limited¹.

Corresponding operator learning models often lack robustness and generalizability, but with a focus on design optimization these are not crucial but only desirable features. It is expected that corresponding approaches will be used only for design exploration but not for design validation. Even if a non-feasible design is suggested during design exploration this will be discovered latest in the validation process and will “only” lead to a delayed time to market and increase development costs.

3.2.1 Laplace operator approximation for manifold learning

One of the most well-studied operators in mathematics, the Laplace(-Beltrami) operator relates the differential structure of a Riemannian manifold to the underlying Riemannian metric. It is usually denoted with the symbol Δ , and is defined purely through the metric tensor field g , so that for a function f on a d -dimensional manifold $(\mathcal{M}, g)$ with curvilinear coordinates $x = (x^1, \dots, x^d)$,

$$\Delta f = \frac{1}{\sqrt{\det g}} \frac{\partial}{\partial x^i} \left(\sqrt{\det g} g^{ij} \frac{\partial}{\partial x^j} \right) f, \quad i, j = 1, \dots, d.$$

In Euclidean space, the metric tensor is the identity everywhere, and so the Laplace-Beltrami operator simplifies to the sum of second derivatives, $\Delta f = \sum_{ii} \frac{\partial^2}{\partial x^i \partial x^i} f$. In scientific machine learning, the relation of the operator to the underlying metric has been exploited for computationally extracting information about the geometry of the manifold [49, 73]. Starting with Laplacian eigenmaps [73, 74], the Diffusion Maps algorithm [49, 50] has been a very popular, efficient,

¹ Datasets for external aerodynamics of cars being a notable example, e.g., [72].

and accurate method for the approximation of the operator and its use in dimensionality reduction [27], construction of smooth basis functions, and manifold learning in general [20, 49–53]. Many extensions have been proposed, even in the direction of a computational extraction of the entire exterior calculus on manifolds [75–78].

3.2.2 Koopman operator framework

The Koopman operator provides a linear perspective on nonlinear dynamical systems by lifting the dynamics into an infinite-dimensional function space [14, 56, 79, 80]. Given a dynamical system $x_{t+1} = F(x_t)$, the Koopman operator $\mathcal{K}$ acts on observables $g(x)$ via composition:

$$\mathcal{K}g(x) = g(F(x)).$$

Although the original system may be nonlinear, $\mathcal{K}$ is a linear operator, making it amenable to spectral analysis and linear approximation techniques.

In SciML, Koopman-based methods aim to learn finite-dimensional approximations of $\mathcal{K}$ from data, often using neural networks or dictionary-based approaches [81–84]. These models can capture long-term dynamics, identify coherent structures, and serve as interpretable surrogates for complex systems. Recent developments include deep Koopman models and extended dynamic mode decomposition (EDMD), which integrate data-driven learning with operator-theoretic insights [83, 85].

Koopman operator learning complements other operator learning methods by emphasizing structure-preserving representations and spectral decompositions [86], particularly useful in systems with oscillatory [87] or quasi-periodic behavior [88]. In a wider scope, algorithms themselves can be considered dynamical systems and analyzed within this framework [89].

For mathematicians, the Koopman framework provides a natural link between nonlinear dynamics and classical spectral theory. By representing nonlinear evolution as a linear operator acting on observables, it opens the door to applying tools from functional analysis, operator semigroups, and spectral decomposition to problems traditionally approached via nonlinear PDE theory. This perspective resonates with well-established ideas in ergodic theory and dynamical systems, offering a mathematically rigorous bridge between data-driven modelling and analytical structure.

3.2.3 Deep Operator Networks (DeepONets)

Based on the universal approximation property of operators [90], DeepONets represent operators using a two-branch neural network: one branch encodes the input function $a(x)$ via point evaluations, and the other branch encodes the evaluation location x . The outputs of both branches are combined to produce the value $u(x)$. This architecture is inspired by the universal approximation theorem for operators and has been shown to generalize well across function spaces [91].

3.2.4 Fourier Neural Operators (FNOs)

FNOs learn operators by representing functions in the Fourier domain and applying learned transformations to their spectral coefficients [47]. This approach is particularly effective for PDEs with translation-invariant structure, such as those arising in fluid dynamics. FNOs can utilize Fast Fourier Transform implementations and are thus computationally efficient, even for input fields of high resolution.

3.2.5 Other architectures

Variants such as Graph Neural Operators, Transformer-based models [92], and kernel-based operator learners have also been explored, each offering different trade-offs in terms of expressivity, efficiency, and interpretability.

Operator learning offers several advantages. Once trained, the learned operator can be evaluated rapidly for new inputs, enabling real-time inference and many-query applications. It avoids the need for discretization during inference, making many algorithms mesh-independent and potentially more scalable. Most recently, operator learning for generalization across varying geometries, boundary conditions, or parameter regimes is being worked on.

Challenges remain, including the need for large and diverse training datasets, the difficulty of incorporating physical constraints, and the theoretical understanding of approximation properties. Nonetheless, operator learning represents a promising direction for accelerating simulations and enabling data-driven discovery in scientific domains.

3.3 Hybrid modeling and surrogate models

In many scientific applications, neither purely mechanistic models nor purely data-driven models are sufficient on their own. Mechanistic models may be incomplete, computationally expensive, or based on simplifying assumptions, while data-driven models may lack physical interpretability or generalization. Hybrid modeling seeks to combine the strengths of both approaches by integrating machine learning components into traditional scientific models. This has been a long-standing issue with several solutions suggested decades ago [93], but has only recently been applied to more challenging problems, and with a broader scope [94].

3.3.1 Hybrid and Gray-box Physics–ML Models

Hybrid Modeling has a long history in the turbulence community [95, 96] and with the emergence of trajectory training [97] has found successful applications in the computer graphics [98] as well as weather/climate prediction. For example Google's NeuralGCM architecture [99], which is effectively a hybrid modeling approach, has outperformed Google Graph Cast approach for weather and climate predictions [100].

A common strategy is to decompose the model into known and unknown components, an approach called “gray-box modeling” [101]. For example, in a dynamical system governed by

$$\frac{du}{dt} = f_{\text{known}}(u) + f_{\text{unknown}}(u),$$

the known part f_{known} is derived from physical principles, while the unknown part f_{unknown} is learned from data using a neural network or other regression model. This approach allows the model to remain grounded in physics while correcting for model bias or missing dynamics. In such gray-box models, machine learning is used to learn unknown components of a partially specified model [101]. For instance, closure terms in turbulence models or constitutive relations in material models can be learned from data while preserving the overall structure of the governing equations [102, 103]. This includes conservation of energy in Hamiltonian dynamics [13].

Most architectures rely on an operator splitting approach to correct the Navier-Stokes Equations, either introducing a local correction directly on the equation level, e.g., [104], or within the solve process, e.g., ([105]; see Fig. 2b). Thereby the correction usually depends on the flow field as well as potentially geometric properties, e.g., represented by a signed distance functions. Architectures are multi-fold and range from CNNs [61] to more involved architectures ([106]; see Fig. 2a).

Accelerating the solve process by using a coarser grid combined with a ML model to capture the effects of unresolved scales, corresponding approaches fall de-facto into the class of multi-scale models [107]. Thus, these models show an impressive generalization capability, but are so far only researched to a limited extend. Open problems include a more rigorous analytical treatment, ideally providing error estimates, and addressing quasi-stationary simulations, as mostly employed in industrial practice².

3.3.2 Surrogate modeling

Surrogate models are computationally efficient approximations of expensive simulations. They are trained to emulate the input-output behavior of a high-fidelity

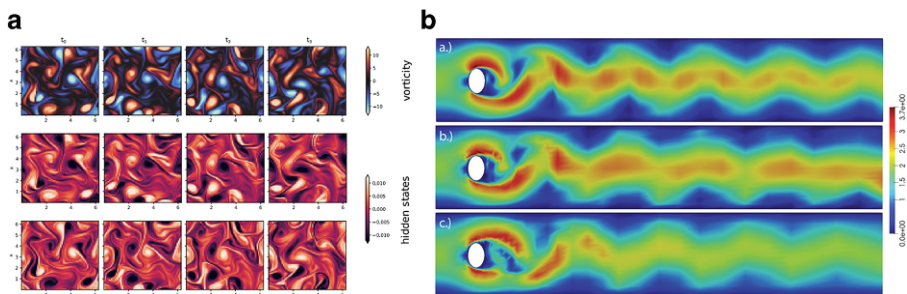


Fig. 2 Prediction of a turbulent velocity field using a hybrid ML model (from [106]) and (from [105])

² Most hybrid approaches in CFD applications addressing only time-dependent flows.

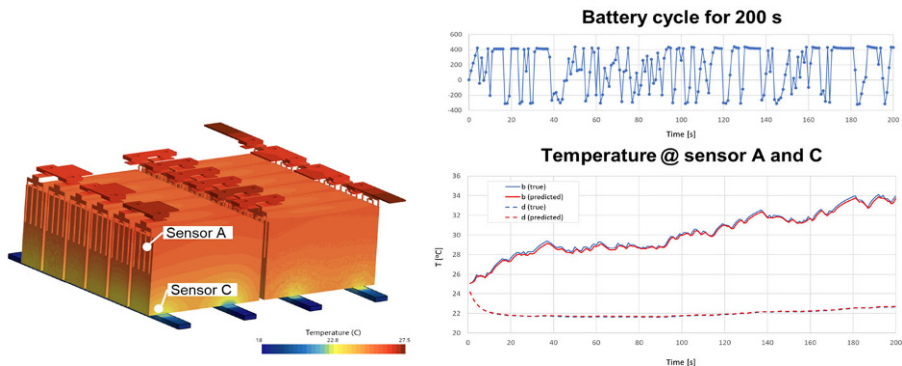


Fig. 3 Prediction of a turbulent velocity field using a hybrid ML model (from [116])

model, enabling rapid evaluation in tasks such as optimization, uncertainty quantification [108], and real-time control. This turns out to be specifically useful if a regularly evaluated subproblem can be replaced, e.g. a constitutive model in an FEM simulation [109]. Common surrogate modeling techniques include:

- **Neural network surrogates:** Deep networks trained on simulation data to approximate the solution map.
- **Reduced-order models (ROMs):** Projection-based methods (e.g., Proper Orthogonal Decomposition) combined with ML to capture dominant modes [110].
- **Emulators:** Gaussian processes or polynomial chaos expansions used to interpolate simulation outputs.

Albeit Operator Learning and Hybrid Modeling have the potential to significantly accelerate predictions times, they are often still too complex to be used as real-time models for decision support in operations ([111]; see Fig. 3), specifically given hardware constraints in edge or embedded computing. Thus, Surrogate Modeling has proven to be a much more versatile technology and specifically reduced order modeling, i.e., the identification of a reduced linear or non-linear subspace in which a model can be formulated more effectively. The latter included projection-based methods such as Proper Orthogonal Decomposition (POD) [112], autoencoders [113], or (Koopman-) Operator Inference (OI) [80]. Also DeepONet [46] with the internal multiplicative decomposition can be considered to fall into this class.

Specifically POD-based methods [114, 115] have demonstrated much success in real-world applications due to their simplicity and robustness [116]. In combination with OI they allow to reduce complex models to a relatively low dimensional set of ODEs [117, 118]. Albeit geometry variations cannot be handled by corresponding approaches, this limitation is usually not relevant since most practical use cases in operation do not involve geometry variations.

While corresponding methods work well for mildly non-linear problems, open challenges remain for more effective and robust methods for non-linear models, e.g., convection cooling and plume formation, the certification/quantification of accuracies, as well the targeted generation of data [119]. If corresponding models are

being used in an operational context, use cases often involve the exploitation of conservative safety margins calling for robust trustful models.

Training and integration Training hybrid models often involves solving coupled optimization problems, where the machine learning component must be trained in conjunction with the numerical solver. Techniques such as adjoint-based gradient computation, differentiable programming, and physics-constrained training are essential for efficient learning.

Advantages and challenges Hybrid models offer several benefits, including improved accuracy by correcting model deficiencies, reduced computational cost via surrogate modeling, and enhanced interpretability and physical consistency. However, they also pose challenges. Ensuring stability and consistency of the coupled system, balancing data-driven and physics-based components, and Designing architectures that respect physical constraints are some of the most crucial ones.

3.4 Random feature methods

Random feature methods provide a scalable alternative to kernel methods and deep neural networks by approximating nonlinear function spaces using randomized basis expansions. Originally introduced to accelerate kernel machines, these techniques have gained renewed interest in Scientific Machine Learning (SciML) due to their simplicity, interpretability, and computational efficiency [37, 120].

The core idea is to approximate a kernel function $k(x, x')$ by an explicit finite-dimensional feature map. For shift-invariant kernels such as the Gaussian RBF kernel, Bochner's theorem implies that the kernel can be expressed as the expectation of a random Fourier feature:

$$k(x, x') = \mathbb{E}_{\omega \sim p(\omega)} [\phi_{\omega}(x) \phi_{\omega}(x')],$$

where $\phi_{\omega}(x) = \cos(\omega^{\top} x + b)$ and $\omega \sim p(\omega)$ is drawn from a distribution related to the Fourier transform of the kernel. In practice, one samples D random frequencies $\{\omega_i\}_{i=1}^D$ and constructs a finite-dimensional feature map $\Phi(x) \in \mathbb{R}^D$, enabling fast linear learning:

$$f(x) \approx w^{\top} \Phi(x), \quad w \in \mathbb{R}^D.$$

In SciML, random feature models can be used to:

- **Accelerate training:** By replacing deep architectures with shallow models in randomized feature spaces, training becomes a convex optimization problem, often solvable in closed form [121].
- **Enable uncertainty quantification:** Bayesian linear regression in random feature space yields tractable posterior distributions, useful for probabilistic modeling.
- **Serve as surrogates:** Random feature models can approximate solution operators or response surfaces with low computational overhead [122].

Variants such as data-dependent sampling [121, 123] improve approximation quality and scalability. Moreover, random features can be integrated into hybrid models or used to initialize neural networks with favorable inductive biases [37, 85, 124, 125]. This includes operator learning [121, 126] and more generalized architectures [85, 125] (recurrent and graph networks). While less expressive than deep networks, random feature methods offer a compelling trade-off between accuracy and efficiency, particularly in low-data or many-query regimes common in scientific applications.

3.5 Probabilistic and Bayesian approaches

Scientific modeling often involves uncertainty—arising from measurement noise, model approximations, or incomplete knowledge of system parameters. Probabilistic approaches provide a principled framework for quantifying and propagating uncertainty, enabling robust predictions and informed decision-making. In the context of Scientific Machine Learning, these methods allow for the integration of data and models in a statistically coherent manner.

3.5.1 Gaussian Processes and kernel methods

Gaussian Processes (GPs) are nonparametric models that define distributions over functions. Given data $\{(x_i, y_i)\}_{i=1}^N$, a GP prior $f \sim \mathcal{GP}(m(x), k(x, x'))$ specifies a mean function $m(x)$ and covariance kernel $k(x, x')$. The posterior distribution incorporates observed data and provides uncertainty estimates for predictions. GPs are particularly useful in low-data regimes and for surrogate modeling [127]. They scale poorly with large datasets if certain aspects of scientific computing are ignored (i.e., using dense matrices with many zeros instead of sparse linear algebra), but with the correct algorithms, they can be very efficient and scalable [128].

3.5.2 Bayesian Neural Networks

Bayesian Neural Networks (BNNs) place distributions over the weights of a neural network, allowing for uncertainty quantification in predictions. Training involves approximating the posterior distribution over weights, typically using variational inference or Monte Carlo methods. BNNs can capture epistemic uncertainty (due to limited data) and are useful in safety-critical applications.

3.5.3 Bayesian inverse problems

In inverse problems, one seeks to infer unknown parameters θ from observations y , given a forward model $y = \mathcal{F}(\theta) + \epsilon$. The Bayesian approach formulates this as a posterior distribution:

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta),$$

where $p(\theta)$ is the prior and $p(y \mid \theta)$ is the likelihood. Machine learning can assist in approximating the forward model $\mathcal{F}$, accelerating inference via learned surrogates or amortized inference techniques. A related approach is so-called Simulation-Based Inference (SBI) [129], which yields distributions for potential solutions of inverse problems and then approximates these distributions with neural networks.

3.5.4 Physics-based priors and constraints.

A key advantage of Bayesian methods in SciML is the ability to incorporate physical knowledge as priors or constraints [130]. For example, conservation laws, symmetries, or known dynamics can be encoded into the prior distribution or used to regularize the likelihood. This improves interpretability and ensures physically plausible inference.

3.5.5 Uncertainty quantification and decision-making

Probabilistic models enable uncertainty-aware predictions, which are essential for tasks such as risk assessment, experimental design, and control. Techniques such as Bayesian optimization, active learning, and ensemble methods can use uncertainty estimates to guide data collection and model refinement. Despite their strengths, probabilistic approaches face challenges in scalability, model specification, and computational cost. Recent advances in probabilistic programming, variational inference, Gaussian process approximations [131], and deep generative models have expanded their applicability in scientific domains.

3.6 Comparison of methods

PINNs are most effective for problems where the governing equations are well known but solution data is scarce, particularly in smooth regimes or with moderate dimensionality. They naturally enforce physical constraints via loss functions, which improves interpretability and plausibility. However, for stiff or highly oscillatory problems, training can be challenging and computationally costly, and current theory offers limited guarantees. Compared with purely data-driven methods, PINNs integrate prior knowledge more directly, but they may underperform reduced-order models when high-fidelity solvers are already fast. Operator learning methods—whether DeepONets, Fourier Neural Operators, or Koopman-based—are particularly valuable in many-query settings, where a solution map must be evaluated repeatedly for varying inputs. DeepONets are versatile but can be data-hungry; FNOs excel for translation-invariant PDEs; Koopman methods shine when the system exhibits coherent structures or oscillatory dynamics and when interpretability is important. While powerful, these methods typically require large, diverse datasets and may struggle to generalize beyond the training distribution. Hybrid modeling is most suitable when reliable partial models exist but certain components are unknown or too expensive to simulate. It offers a balance between physical fidelity and data-adaptiveness. Gray-box approaches are attractive for preserving physical structure while correcting bias, but their stability depends on careful coupling between the

physics and ML parts. Surrogate modeling, especially reduced-order approaches, is ideal for real-time or embedded applications with fixed geometry and moderate nonlinearity. In contrast, purely data-driven models may be faster to train but risk violating physical constraints. Random feature methods are attractive when computational speed, simplicity, and interpretability are priorities—particularly in low-data or many-query regimes. They trade some expressivity compared with deep architectures for the benefit of convex training and analytical tractability. This makes them well-suited for uncertainty quantification and for serving as lightweight surrogates in larger hybrid frameworks. Probabilistic methods are the natural choice when quantifying uncertainty is as important as making predictions, especially in safety-critical or decision-making contexts. Gaussian processes excel in low-data regimes with smooth solution spaces; Bayesian neural networks handle more complex patterns but at greater computational cost. These methods integrate naturally with inverse problems and data assimilation but require careful prior design and face scalability challenges in high dimensions.

4 Applications across scientific domains

4.1 Fluid dynamics and turbulence modeling

CFD is among the computationally most demanding applications in industrial practice. Corresponding industrial models have frequently up to 10^8 – 10^{10} degrees of freedom and take days to solve. At the same time exploding system complexity requires fast predictions to better exploit design options (design exploration) as well as real-time predictive models enabling a more more efficient operation (operation support). Scientific Machine Learning (SciML) is a key enabler to realize corresponding fast models, potentially even real-time. Thus it is a key technology for Digital Twins [132–134]. To achieve such fast predictive models in the context of CFD Operator Learning (Sect. 3.2), Hybrid Modeling (Sect. 3.3.1), and Surrogate Modeling (Sect. 3.3.2) have demonstrated to be effective approaches.

4.2 Complexity handling and AI-supported synthesis of automation and control architectures

AI-based simulation of physical systems and AI-supported generation of executable automation and real-time control software can be incorporated into a single framework. The associated mathematical requirements comprise the automated AI-based design of validated simulation models [135], appropriate control structures and related functions as well as their calibration. These aspects face respective software challenges concerning the validation, verification and release of the subsequent control algorithms. The step from simple individual control modules [136] to a system-wide automation and control perspective offers an enormous potential in industrial applications and time to market, while strengthening the competitiveness of the industry.

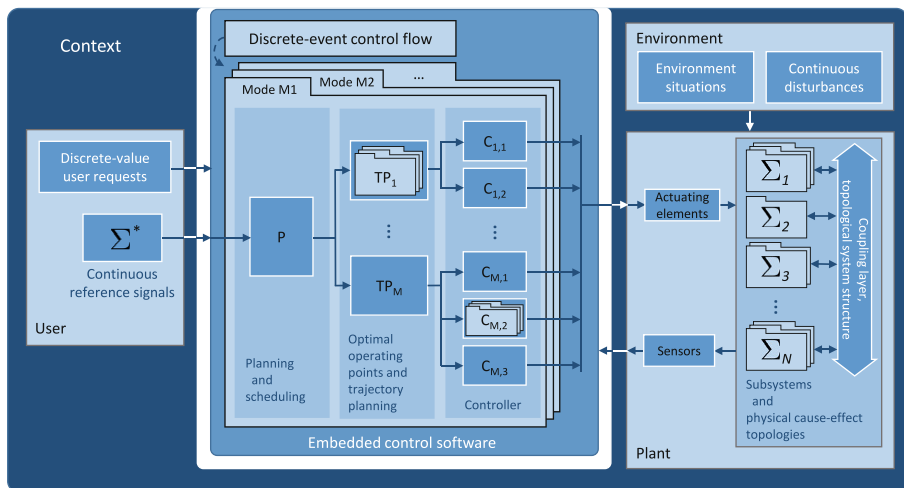


Fig. 4 Representation of a cascaded block diagram structure with hierarchical planning coordinators for the management of networked (sub-)systems. (This graphics has previously been published in [139] under Creative Commons Attribution 4.0 Public License.)

A system-wide automation perspective requires the consideration of multifaceted qualities, such as multi-physics and multi-time scale dynamics [137], mixed-physics—as well as data-based models, networked and modular hierarchical control systems, federated-learning approaches, as well as optimal operational strategies [138]. Such modular hierarchical approaches [139] lead to cascaded structures with 1-to- $\{n\}$ relationships, i.e., with n coordinated subsystems, see Fig. 4. The synchronisation of constrained modules and potentially entangled constraints over the hierarchical layers shows respective coordination needs within the overall control structure whereat the complexity occurring on component level may already be noteworthy, e.g. [140, 141]. On the software side, these qualities need to be complemented with measures supporting a respective productivity increase, such as correct-by-construction approaches emphasizing on consistency [142], reusability and maintainability [143, 144]. A stringent development process combined with an automatization of verification tests further completes the view.

The potential of AI-supported design and integration, enabling the systematic interconnection of simple control elements for a particular application is likely to be a key enabler for complex system design and for fast operational implementations. The approach requires a co-design view between a mathematics-oriented functional control perspective [139] and a software- and implementation-oriented coding perspective [143, 144].

4.3 Holistic energy management for vehicles with different drivetrain topologies

Sustainability in mobility requires modern methods to realise optimal operational strategies for vehicles and vehicle fleets that minimise their CO₂ emissions and

energy consumption. These requirements apply for all vehicle classes and drivetrain topologies [145], i.e., electric powertrains, powertrains with fuel cells, as well as hybrid drivetrains with internal combustion engines and batteries [146]. In the case of electric vehicles, the energy demand for comfort functions such as cabin heating directly influences the possible range [147, 148]. Furthermore, waste heat recovery is used to improve efficiency [149].

The foundations of these methods are physics-based or data-based mathematical models, or hybrid modelling approaches. Their industrial use demands both robustness and trustworthiness in connection with an efficient parameterisation [150–152]. Control-oriented modelling in combination with model order reduction are often a prerequisite to accomplish real-time operation due to resource constraints of embedded control units. Future centralised vehicle computers will provide additional computation power and enable increasingly complex algorithms, e.g. optimisation on system level. An external communication and computer infrastructure [153] offers further possibilities for connected services such as federated learning or networked optimal control.

4.4 Towards a circular CO₂ ecosystem for sustainable mobility

The environmental impact of mobility and its related energy demand is an important aspect of the overall energy consumption of societies. Therefore, it is desirable to minimize the energy and CO₂ emissions of the mobility sector as far as possible. This endeavor requires an increasing inclusion of renewables such as wind and solar in the energy supply. A key aspect are modern storage technologies for electrical energy in order to bridge the intermittent and decentralized availability of renewables. In order to provide and balance the requested present energy demand, the storage possibilities have to be combined with optimal energy management concepts [154, 155]. This concerns different time- and application-scales. I.e., from family housing equipped with e.g. solar modules, batteries for local short-time storage, and charging of electric vehicles, towards office buildings, factories and production sites. For long-term as well as large-scale storage, the possibilities and favorable energy densities of adequate chemicals are an additional option. Such concepts would then include approaches for an alternating production, storage, and repowering of these chemicals [156–158].

In order to achieve not only a CO₂ neutral operation of electric vehicles but also of internal combustion powertrains (including hybrids), tailor-made e-fuels [159] are an additional promising solution option. Their production [160, 161] will be dependent on the intermittent renewable energy availability as described above. The fluctuations in the energy supply might further influence the production process operation due to cost reasons. E.g., instead of using large energy storage to smooth the energy supply, either the production quantity or the product quality can be adjusted to the given energy availability. Therefore producers will offer tailor-made fuels in different qualities and quantities [162]. Based on digital fuel information and scientific machine learning approaches [159], an emission-minimal combustion [163, 164] needs to be guaranteed within such a circular CO₂ ecosystem.

5 Open problems, outlook, and discussion

Despite significant progress, scientific machine learning faces several open challenges that limit its broader adoption and reliability. Key issues include the lack of theoretical guarantees for generalization, especially in high-dimensional and data-scarce regimes, and the difficulty of integrating domain knowledge in a principled and scalable way. Moreover, ensuring physical consistency, interpretability, and robustness under distributional shifts also remains a central concern, particularly in safety-critical applications.

Looking ahead, the future of SciML lies in the development of hybrid frameworks that seamlessly combine data-driven models with first-principles physics, enabling more accurate and generalizable solutions. Advances in uncertainty quantification, adaptive sampling, and transfer learning will be crucial for extending SciML to real-world scientific problems where data is limited or expensive to obtain. Additionally, the co-design of algorithms and hardware, such as neuromorphic computing and specialized accelerators, may unlock new levels of efficiency and scalability. For domains where large data sets are available, scalable and generalizable architectures like neural operators will help to condense and operationalize these data. Independent of the specific application, it will continue to be important to focus on the problems and challenges that must be solved, rather than trying to incorporate “AI into everything”.

Finally, interdisciplinary collaboration will be essential to drive innovation in SciML. Bridging the gap between machine learning researchers and domain scientists can foster the creation of models that are not only powerful but also trustworthy and aligned with scientific understanding. As the field matures, establishing benchmarks, open datasets, and standardized evaluation protocols will be critical for measuring progress and guiding future research directions.

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